NETWORKING ESSENTIALS • DAY 1 OF 6 • 4 HOURS

Day 01 — Foundations

How Do the Networks Talk?

From the Industry to the Classroom — Building the IT's Future, Today and Together!

Cyber.SoHo Educational Hub

Welcome & Today's Goal

Why Day 1 Matters

- You will leave today able to describe a network in plain English
- You will have a working lab environment on your laptop
- You will have produced your first real engineering artifact — a 1-page network plan

No prior networking knowledge required. We start at zero.

Today's Schedule (4 hours)

Plus 1 h home work per exercise for the report.

Block	Duration	Content
Lecture 1	1 h	What is a network? Types & speed
Exercise 1	1 h	Lab Tools Tour — install & verify
Lecture 2	1 h	Getting online + documentation
Exercise 2	1 h	Project Stage 1 — Acme SoHo

Learning Outcomes

By the end of today you will be able to:

1. Define what a network is and what it carries
2. Distinguish PAN, LAN, MAN, WAN
3. Read common speed/capacity units (bps → Gbps)
4. Run a basic lab on VirtualBox + GNS3
5. Plan a small office network on paper

The Project — Acme SoHo Consulting

Your Recurring Client for 6 Days

- Fictional 8-person consulting firm
- 2-floor new office, moving in 4 weeks
- Each day = 1 stage of the project
- By Day 6 → complete network plan

Cyber.SoHo Network Build Challenge — your portfolio piece.

LECTURE 1

What, Which, How Fast?

Networks defined, classified, and measured

Lecture 1 Roadmap

Three Big Questions

1. What is a network? (and what actually moves through it?)
2. Which kinds exist? (PAN, LAN, MAN, WAN)
3. How fast / how much? (bandwidth, throughput, latency)

Each question answered with a real-life example you already know.

1.1 What Is a Network?

WORKING DEFINITION

A network is two or more devices connected by a medium that lets them exchange data using agreed rules.

Three essential ingredients:

- **Devices** — endpoints that produce/consume data
- **Medium** — copper, fibre, or radio waves
- **Rules** (protocols) — how they take turns talking

What Counts as a 'Device'?

Endpoints make the data. Intermediate devices move it.

Endpoints	Intermediate Devices
Laptop / desktop	Switch
Phone / tablet	Router
Server / NAS	Access Point (AP)
Printer / Smart-TV	Firewall
IoT (sensor, lock, doorbell)	Modem

What Carries the Data?

Three Physical Media

1. Copper — twisted-pair (Ethernet), coaxial
2. Fibre optics — pulses of light through glass
3. Radio waves — Wi-Fi, Bluetooth, cellular

All three carry the same logical data — only the physical layer differs.

Why 'Protocols' Matter

Rules of the Road

- Devices speak dozens of protocols simultaneously
- Each protocol = one job (addressing, error-checking, naming…)
- Without protocols → digital chaos
- Standardised by IETF, IEEE, ISO

Protocol = the grammar of the network.

Key Takeaway — What Is a Network?

🔑 **KEY TAKEAWAY**

A network = devices + medium + rules.

Strip any one of those three and you have no network — just hardware in a box.

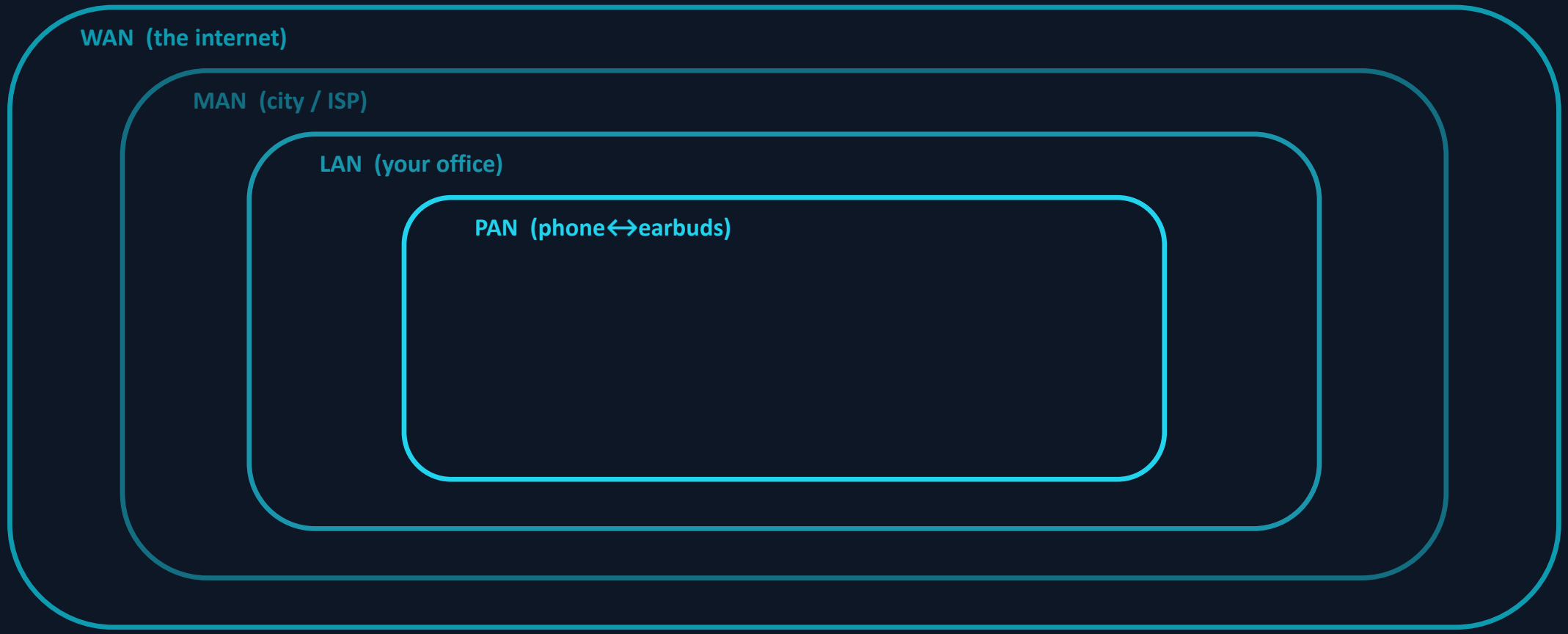
1.2 Types of Networks (1/2)

Sized by geographic reach.

Type	Reach	Typical Use
PAN — Personal Area	~1 m	Phone ↔ earbuds (Bluetooth)
LAN — Local Area	Building	Home / office Wi-Fi & Ethernet
MAN — Metropolitan	City	Campus, ISP municipal fibre
WAN — Wide Area	Country / globe	The internet itself

1.2 Types of Networks (2/2)

How They Nest



Smaller networks plug into larger ones — like Russian dolls.

Other Network Flavours You'll Hear

Function-based, not size-based

- **WLAN** — Wireless LAN (Wi-Fi)
- **SAN** — Storage Area Network (data-centre disks)
- **CAN** — Campus Area Network (university)
- **VPN** — Virtual Private Network (overlay tunnel)

Same building blocks, different purpose.

Key Takeaway — Network Types

🔑 **KEY TAKEAWAY**

Pick the network type by distance + purpose:

meters → PAN • building → LAN • city → MAN • world → WAN

1.3 Speed & Capacity — Three Words

Stop Confusing Them

- **Bandwidth** — theoretical maximum capacity
- **Throughput** — actual data delivered per second
- **Latency** — delay before the first bit arrives

Bandwidth is the size of the pipe. Throughput is what comes out. Latency is how long until water reaches you.

The Highway Analogy

Visualise it.

Network term	Highway equivalent
Bandwidth	Number of lanes
Throughput	Cars actually arriving per minute
Latency	Travel time from on-ramp to exit
Jitter	Variation in arrival times
Packet loss	Cars that crashed and never arrived

Units of Capacity

Read the lowercase 'b' carefully — 8 bits = 1 byte.

Unit	Meaning	Order of magnitude
bps	bits per second	base unit
Kbps	kilobits / s	10^3
Mbps	megabits / s	10^6
Gbps	gigabits / s	10^9
Tbps	terabits / s	10^{12}

Real-World Reference Speeds

Memorise These Anchors

- 2 Mbps — basic web + email
- 25 Mbps — single 4K stream
- 100 Mbps — typical home fibre
- 1 Gbps — modern home/SoHo fibre
- 10 Gbps+ — business / data-centre

Uplink (upload) is often much lower than downlink — check before video-conferencing.

Speed Sanity Formula

Estimate download time

$$\text{Time (s)} \approx \text{File size (Mb)} \div \text{Throughput (Mbps)}$$

Quick mental check:

- 1 GB file = 8000 Mb
- On 100 Mbps connection → 80 s ideal
- Reality: ~50 % overhead → expect 2 minutes

Latency Reference Numbers

What 'Fast' Actually Means

- < 1 ms — same LAN
- 5–20 ms — same city / ISP
- 40–80 ms — same continent
- 150–300 ms — intercontinental
- 600+ ms — satellite uplink

Latency is bound by physics. Bandwidth is bound by money.

Key Takeaway — Speed & Capacity

🔑 **KEY TAKEAWAY**

Bandwidth ≠ Throughput ≠ Latency.

A '1 Gbps' pipe can still feel slow if latency is high or packet loss is wrecking throughput.

Lecture 1 Recap

Three Things to Remember

1. Network = devices + medium + protocols
2. PAN < LAN < MAN < WAN — sized by reach
3. Bandwidth, throughput, latency are different — never trade names

Ready for Exercise 1 → install & verify your lab tools.

EXERCISE 1

Lab Tools Tour

Install & verify — your full free industry stack

Exercise 1 — Overview

Goal: Install & Verify Your Lab

You will install and prove-it-works for:

1. NetAcad account (Cisco free training portal)
2. VT-x / AMD-V virtualisation enabled in BIOS
3. VirtualBox (Type-2 hypervisor)
4. GNS3 + GNS3 VM (network simulator)
5. Cisco Modeling Labs Free (5-node free tier)

Why These Specific Tools?

Industry Stack — Rationale

- **VirtualBox** → free, cross-platform, GPL v3
- **GNS3** → real Cisco / Juniper images, free
- **CML** Free → official Cisco simulator, 5-node lab
- **NetAcad** → Cisco's free curriculum + Packet Tracer

Total cost: €0. Total industry credibility: high.

Exercise 1 — Deliverable

Your Report Must Show

- Screenshot of NetAcad enrolment confirmation
- BIOS screenshot (or systeminfo) proving VT-x / AMD-V is ON
- VirtualBox launching a test VM
- GNS3 + GNS3 VM green status
- Cisco CML Free login success page

Detailed enough that a beginner could follow it.

LECTURE 2

Getting Online & Writing It Down

The four pieces — and the doc-pack that outlives everyone

Lecture 2 Roadmap

Two Topics

1. Getting online — what every device needs
2. Documentation & schemas — why and how to draw

The first half is technical. The second half is what separates juniors from seniors.

2.1 Four Pieces Every Device Needs

To Reach the Internet

1. IP address (its identity on the network)
2. Subnet mask (which neighbours are local)
3. Default gateway (the door to outside)
4. DNS server (name → address phonebook)

Miss any one → 'I have Wi-Fi but no internet.'

IP Address — The Postal Address

Both can coexist (dual-stack). You'll see both in production.

	IPv4	IPv6
Length	32 bits	128 bits
Format	192.168.1.10	2001:db8::1
Total addresses	~4.3 billion	~3.4×10^{38}
Status	Running out	Future-proof

Subnet Mask — The Neighbourhood Boundary

What It Decides

- Tells your device which IPs are local
- Local IPs → talk directly
- Non-local IPs → send via gateway

Common masks:

- 255.255.255.0 (/24) → 254 usable hosts
- 255.255.0.0 (/16) → 65 534 hosts
- 255.255.255.128 (/25) → 126 hosts

Default Gateway — The Front Door

Always Required

- The router's IP on your subnet
- Anything off-LAN travels through it
- Without it → device is island-locked

Think of it as the post-office for everything outside your street.

DNS — The Internet Phonebook

Names → Addresses

- Humans use names (google.com)
- Machines use IPs (142.250.190.46)
- DNS is the translator

Public resolvers worth knowing:

- 1.1.1.1 — Cloudflare
- 8.8.8.8 — Google
- 9.9.9.9 — Quad9 (privacy / security focus)

Two Ways to Get These 4 Pieces

Modern home/SoHo = DHCP for clients, static for infra.

	Static	DHCP (Dynamic)
Set by	Human, manually	Router, automatically
Used for	Servers, printers, gateways	Laptops, phones, IoT
Pros	Predictable	Zero-touch
Cons	Tedious; error-prone	Address can change

DHCP — The Four-Step Handshake

How 'It Just Works'

1. Discover — 'Anyone there? I need an IP.'
2. Offer — 'Here, take 192.168.1.42.'
3. Request — 'Yes, I'll take it.'
4. Acknowledge — 'Confirmed — it's yours for 24 h.'

Mnemonic: D - O - R - A

Key Takeaway — Getting Online

🔑 **KEY TAKEAWAY**

IP + Mask + Gateway + DNS = Online.

Drop any one and you'll feel 'the Wi-Fi is down' — even though the cable is fine.

2.2 Documentation — The Boring Superpower

Why It Matters

- Networks outlive the engineers who build them
- The job at 3 a.m. is fixing, not rediscovering
- Auditors, insurers, and lawyers want paper
- Every senior I respect documents first

No diagram = no network.

Logical vs Physical Diagrams

Real environments need both — they answer different questions.

Logical	Physical
Subnets, VLANs, routes	Cables, ports, racks
Audience: engineers	Audience: technicians
'What talks to what'	'What plugs into what'
Stable across moves	Changes on every move

The Five Documentation Artifacts

Minimum Viable Network Doc Pack

1. Logical diagram — VLANs, IP ranges, routing
2. Physical diagram — racks, cabling, ports
3. Device inventory — make, model, MAC, IP, location
4. IP plan — which subnet for which purpose
5. Change log — every modification, dated

Skip any of these and future-you will hate present-you.

Tools of the Trade

Free / Open-Source First

- **draw.io** / diagrams.net — diagramming
- **Excalidraw** — hand-drawn-look sketches
- **LibreOffice** Calc — inventory & IP plan
- **Netbox** / phpIPAM — IP address management
- **BookStack** — wiki / knowledge base

Paid tools (Visio, Lucidchart, SolarWinds) are nice — not necessary.

Naming Conventions

Boring but Critical

- **Devices:** SITE-ROLE-NUMBER → MTL-SW-01
- **VLANs:** purpose-based → VLAN10_CORP
- **Subnets:** document the 'why', not just the IP
- **Files:** dated → topology_v1_2026-04-28.drawio

Consistency > cleverness. Future-you needs to grep this.

Key Takeaway — Documentation

🔑 **KEY TAKEAWAY**

The diagram is the network.

If it's not drawn and named, it does not exist on Monday morning when somebody calls in sick.

EXERCISE 2

Project Stage 1 — Acme SoHo

Discover & Plan — translate a vague brief into a real plan

Exercise 2 — The Brief

Acme SoHo Consulting

- 8 employees on 2 floors
- Laptops, phones, NAS, MFP, video-conf TV, IoT
- 1 Gbps fibre uplink
- 2 SSIDs required (corporate + guest)
- Boss: "We deal with client financial data — do not embarrass me."
- Deadline: 4 weeks

Exercise 2 — Your Two Artifacts

Deliverables for Stage 1

1. Acme_SoHo_Requirements_v1.md — 1-page requirements doc
2. Acme_SoHo_Topology_v1.drawio — first-pass logical sketch

Plus:

- My_Reflection.md (5 lines)
- Day_01_Exercise_02_Report.md (full step-by-step report)

Exercise 2 — Topology Must-Haves

Minimum Elements to Draw

- Internet cloud (top)
- Edge router / firewall
- Core switch (server closet)
- 1 access switch per floor
- 2 APs (one per floor)
- Endpoint groups: 8 laptops, MFP, NAS, TV, IoT
- 4 VLAN colours (Corporate / Servers / Guest / IoT)

Exercise 2 — VLAN Plan

Recommended first-pass segmentation. Each VLAN = different colour in your diagram.

VLAN	ID	Purpose
Corporate	10	Laptops + corp phones
Servers	20	NAS, MFP, video-conf TV
Guest	30	Guest Wi-Fi (isolated)
IoT	40	Smart-lock, doorbell, sensors

Thank You.

Questions? Stay 5 minutes after class.

Tomorrow — Day 02: Cabling, Standards & Reference Models.

Have VirtualBox + GNS3 already running.

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